

Anthony L. Brown

AI Application Developer | Field Service Engineer | Electro-Mechanical Systems Technician
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PROFESSIONAL SUMMARY

I am an experienced semiconductor, electro-mechanical technician, and aspiring AI Application Developer. For my most recent project, I improvised tooling, abstracted standards, and intuited architectural decisions that AI implemented. Drawing from my semiconductor experience, I have invented an original tactical-visual trading methodology and an accompanying governance harness for AI co-development that handles indefinite-horizon session state, a layered rule discipline, and an administrative control surface (LOCK / UNLOCK / SUSPEND semantics). Previously, I developed semiconductor manufacturing procedures and trained fellow technicians. I am able to translate abstract concepts across domain boundaries — semiconductor metrology to governance indexing, Statistical Process Control to algorithmic trading rules. 22+ years of precision electro-mechanical systems experience (U.S. Navy AEGIS; 12-year semiconductor CMP) underpins a hands-on troubleshooting and root-cause discipline that transfers cleanly to AI-orchestrated software work.

CORE COMPETENCIES

■ AI Project Direction & Governance	■ Tactical & Visual Trading Strategy Design	■ Documentation Standards Intuition (IEEE / PMI)
■ AI Governance Harness Design (Inventor — SADP)	■ AI-Orchestrated Software Deployment	■ Systems Architecture Intuition
■ 34-Rule Governance Framework Design	■ Root Cause Analysis	■ Cross-Domain Translation
■ Simulation Improvisation	■ Electro-Mechanical Systems	■ Algorithmic Trading Strategy Comparison

PROFESSIONAL EXPERIENCE

AI Application Developer — Independent Trading Platform

Independent | Portland, OR | 2024 – Present

- Invented the **Harvest-Fold Cycle**: a tactical, visually-derived accumulation methodology that converts price oscillation into asset accumulation without forecasting direction. Validated across 13 assets and 3 market regimes in a controlled 39-simulation battery; corroborated by 50 days of live trading on the deployed platform (2,000+ executed trades).
- Invented **SADP (Structured AI Development Protocol)**: a three-component governance harness for AI co-development, in the same family as community-developed harnesses now formalizing across the ecosystem — Anthropic's handoff-file pattern for long-running agents, AGENTS.md-style project-rule files (adopted by 60,000+ repositories), the Cavekit / Cavemem session-memory approach, and the broader harness-engineering discipline of context resets, structured handoff artifacts, and phase gates. SADP differentiates by combining all three concerns in a single methodology: an atomic generational handoff (preserving session state across context boundaries), a 34-rule discipline layer organized into nine functional groups, and an administrative control surface with LOCK / UNLOCK / SUSPEND semantics over the rule set — designed end-to-end from operating experience rather than from a literature survey.
- **Translated abstract domain knowledge across boundaries**: applied semiconductor metrology principles to design a reliability-targeting index inside the governance framework; applied Statistical Process Control concepts to inspire the invention of Grid Trading Bots; translated a visual trading methodology into a logical, enforceable rule set that AI then implemented in software.

- **Improved RAIntSimBat** — a Monte Carlo simulation harness — to validate the translation from visual methodology to enforceable governance rules. Designed the methodology and verification protocol independently; AI implemented the simulator code under direction. Refined the testing discipline iteratively by having Claude teach the appropriate test types as platform complexity grew.
- **Directed AI deployment** of the resulting platform into live operation against operator capital. Live trading discipline patterns are consistent with the simulated baseline across the 50-day evaluation window.

Field Service Engineer III+

WGNSTAR (contracted to Intel Corporation) | Portland, OR

- Decontamination and decommissioning of Applied Materials CMP polishers (LK, LK Reflexion, LK Prime) at Intel Ronler Acres.

Independent Contractor

Self-Employed | Asheville, NC | July 2020 – 2024

- Independent study in trading methodology design, statistical simulation, and AI-orchestrated software architecture, culminating in the trading platform and governance harness now in active development.

Customer Engineer III

Applied Materials | Chandler, AZ | September 2019 – May 2020

- Installation and qualification of CMP 300mm semiconductor manufacturing equipment.

Field Service Engineer II

Axus Technology | Chandler, AZ | June 2018 – July 2019

- Refurbishment, repair, and upgrade integration of legacy 200mm CMP equipment (AMAT MIRRA, IPEC 472, Strassbaugh 6DSSP).

Senior Equipment Technician

IM Flash Technologies LLC | Lehi, UT | September 2006 – April 2018

- 12-year tenure maintaining AMAT LK Reflexion / Prime, EBARA 300S / 300X, NOVA 3090 / 3090NXT for NAND flash semiconductor manufacturing.
- Deep expertise in root cause analysis, cross-discipline troubleshooting, and technical documentation for mission-critical equipment.

Firecontrolman 2nd Class (FC2)

United States Navy | Multiple Locations | August 1996 – April 2005

- Operated, maintained, and repaired CIWS Phalanx and AEGIS Combat System on active-duty vessels over 8.5-year service career.

EDUCATION & TRAINING

U.S. Navy: Tech Core (1997) ■ Fire Controlman A School ■ Automatic Gunfire Control Systems (1998) ■ AEGIS Computer Operation & Maintenance (2003)

IM Flash Technologies: AMAT LK / EBARA 300S Maintenance (2006–2018) ■ NOVA 3090 / 3090NXT Recipe Maintenance (2008–2018)

Independent Study: Trading Methodology Design ■ Technical Analysis ■ Statistical Simulation (improvised) ■ AI Co-Development Workflows

INVENTIONS & INNOVATIONS (PATENT PENDING)

- **Harvest-Fold Cycle Accumulation Strategy** — structural guarantee: every fold accumulates more asset than the preceding scrum sold.

- **Structured AI Development Protocol (SADP)** — three-component AI co-development governance harness, applicable across any software domain.
- **Compound Profit Folding** — exponential target growth via fold-profit reinvestment.
- **Band Travel Detection** — volatility-normalised trade trigger using Bollinger Band width.
- **Position-Aware Technical Analysis** — context-dependent indicator reinterpretation based on Bollinger Band position.
- **Landing Strip Detection** — original consolidation pattern adapted from semiconductor metrology.